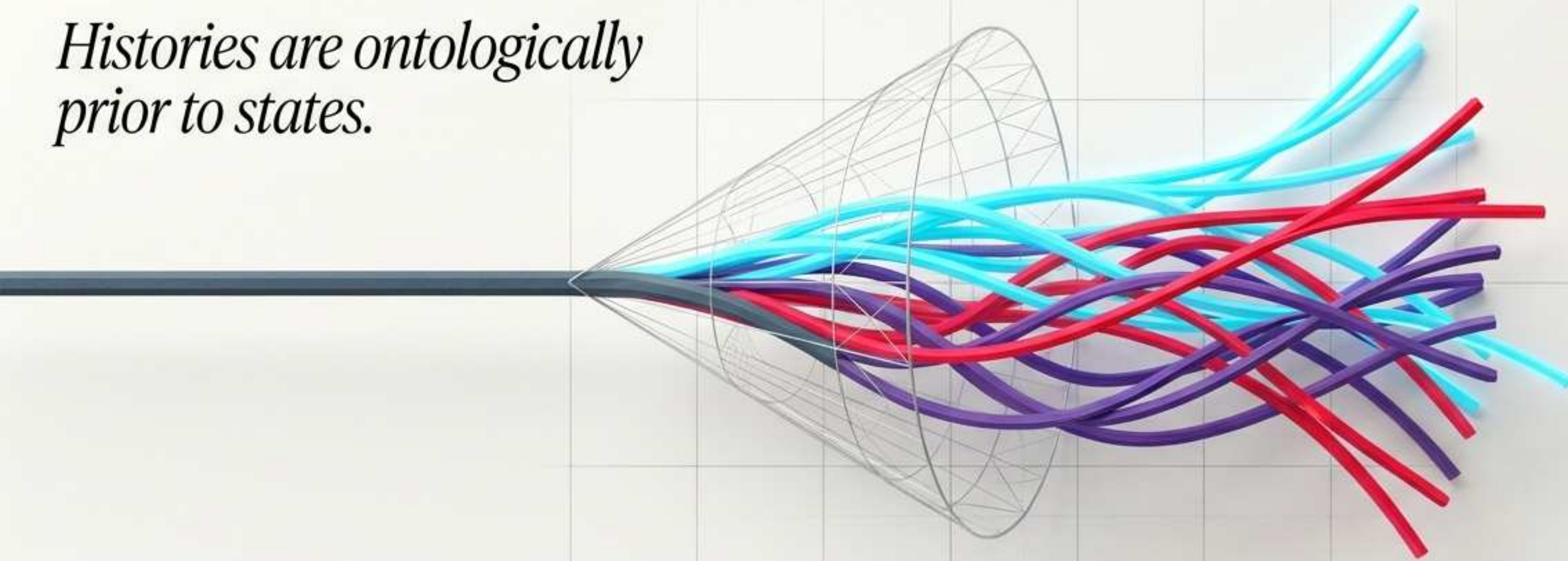


# Spherepop: The Algebra of Reachable Computation

*Histories are ontologically prior to states.*



**Primary takeaway:**

Computation is not the manipulation of present values; it is the progressive restriction of future possibility.

**Context:**

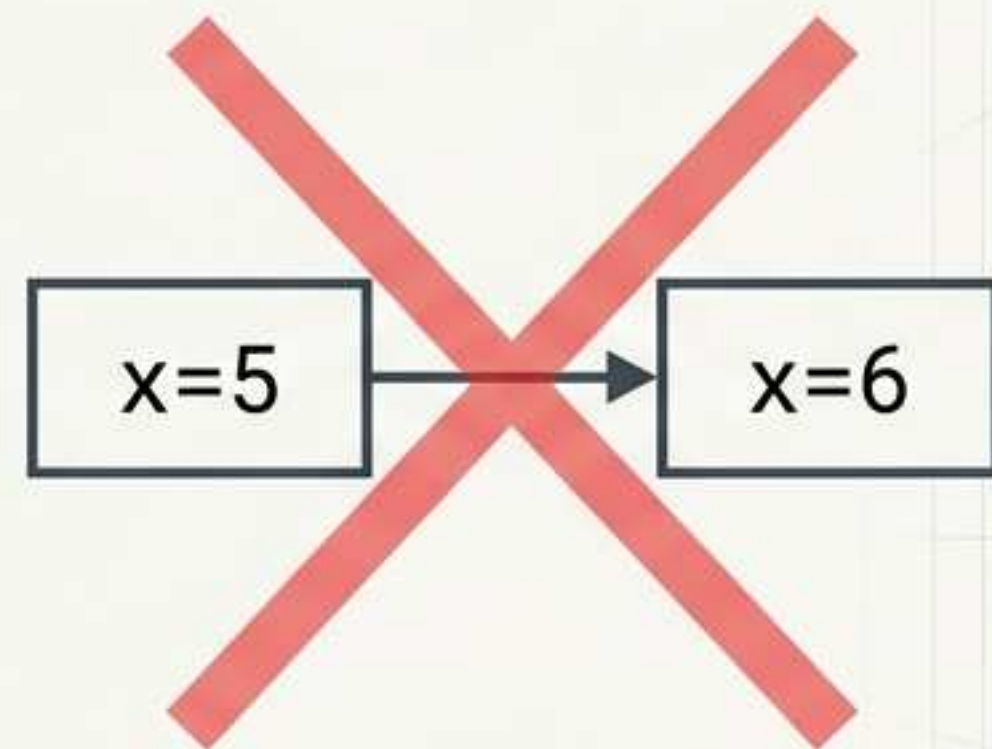
A note is an externalized continuation.

# The Illusions of State and Memory



## The Passive Theory of Notes

A grocery list does not record a past event;  
it constructs a future action.

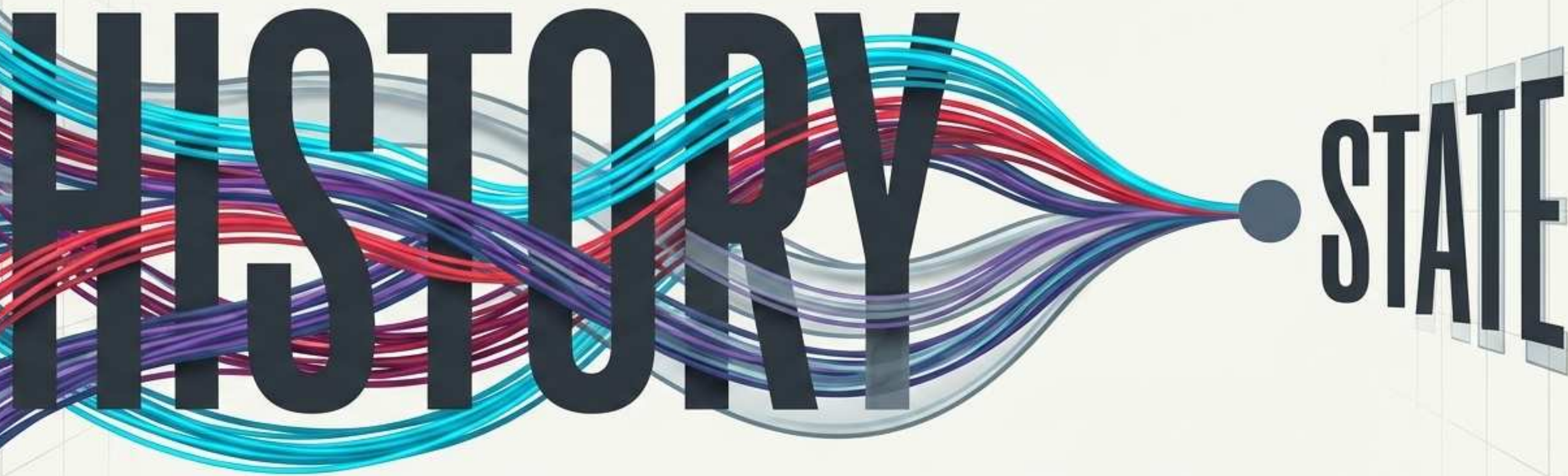


## The State-Centric Assumption

A final value erases the sequence of  
decisions that produced it.

Both paradigms make the same error: they mistake the terminal summary for the active substance. They value the snapshot while discarding the trajectory.

# Histories Precede States



State is a derived notion—a collapsed residue of history.

A state is only instrumentally useful because it is a node in a navigable graph of continuations.

To prevent the loss of history, computation must treat the trajectory as its fundamental object.

# A New Diagnostic for Computation

## State-Centric Assumption

## The Spherepop Inversion

Current State

Primary Object

Entire **Trajectory** (**History**)

Named locations holding values

Variables

Identifiers tied to historical **commitment** events

Race conditions & Use-after-free

Bugs

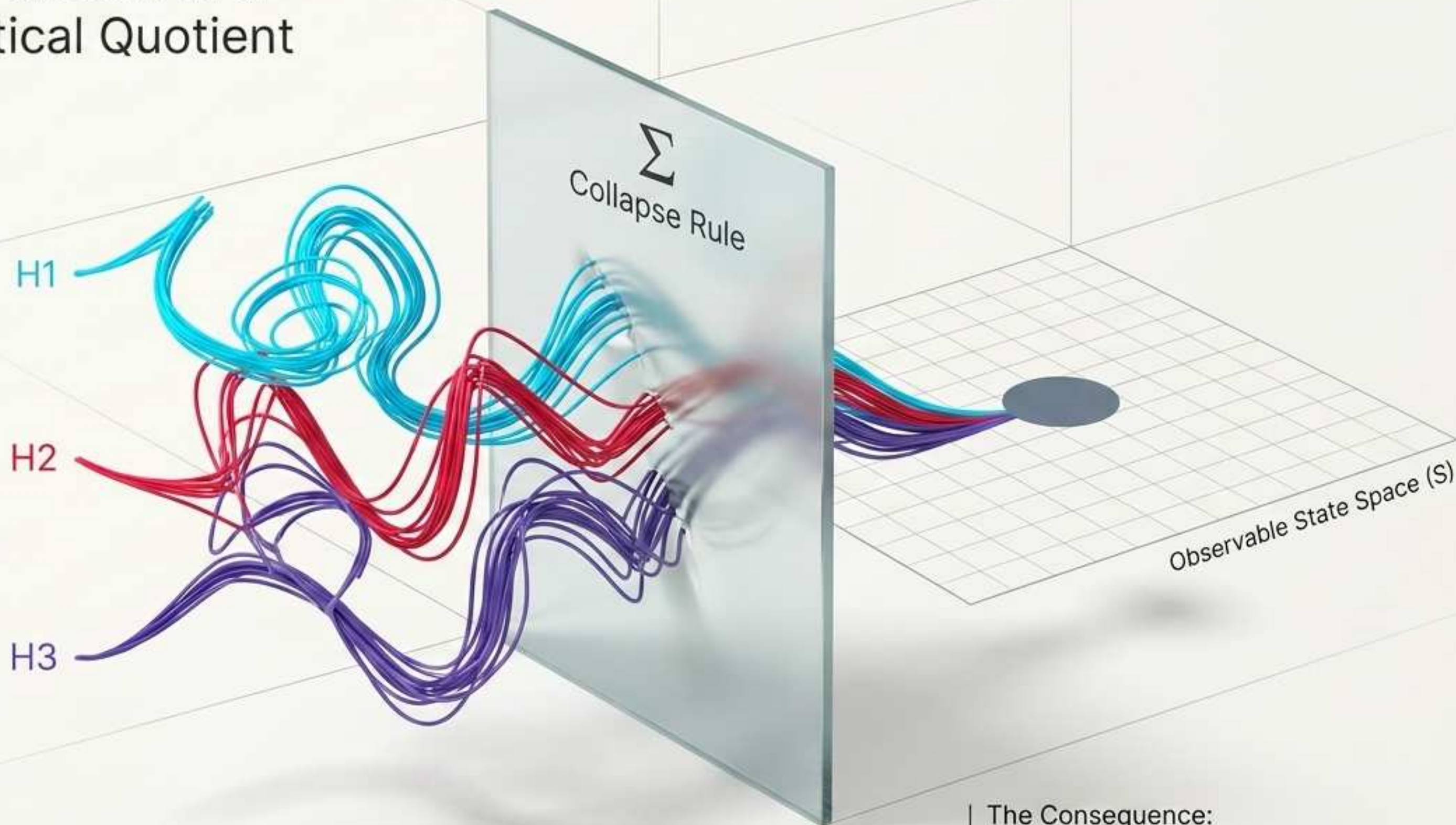
Admissibility certificate **failures**

Machine State manipulation

Compiler Target

**Event** Intermediate Representation (IR)

# The State Illusion is a Mathematical Quotient



## The Proof:

For any non-trivial state space, multiple distinct histories map to the identical observable state.

## The Consequence:

Any program exposing only the collapsed state to its user has silently discarded distinguishing information.

# The Four Operators of Possibility Space

**Pop**  
*Commitment*  
Commits to a specific value,  
permanently narrowing  
the Option Space.



**Refuse**  
*Documented Inadmissibility*  
Excises a path from the  
Admissibility set and permanently  
embeds the reason.



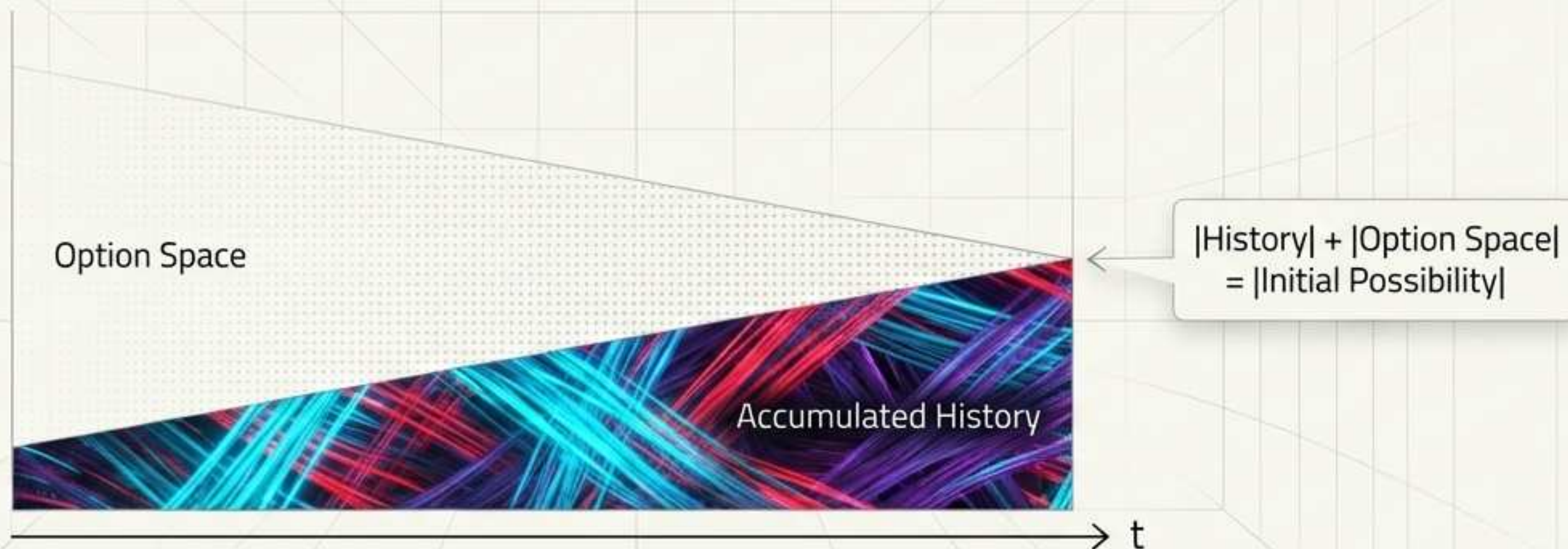
**Bind**  
*Dependency*  
Creates a structural link  
between two trajectory elements  
without consuming either.



**Collapse**  
*Observation*  
Projects a history into an observable  
state under a specific rule. Requires  
an admissibility certificate.



# The Conservation Law of Possibility



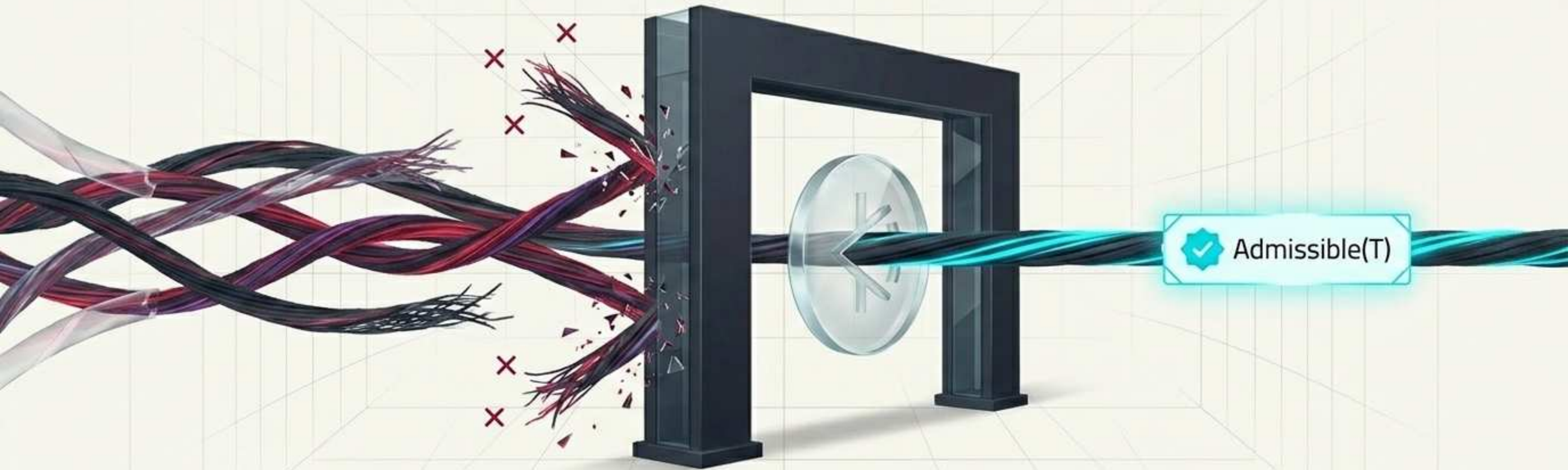
Possibility is never created or pop.

Possibility is never created or destroyed. It is either consumed by Pop or documented as inadmissible by Refuse.

Execution is strictly irreversible.

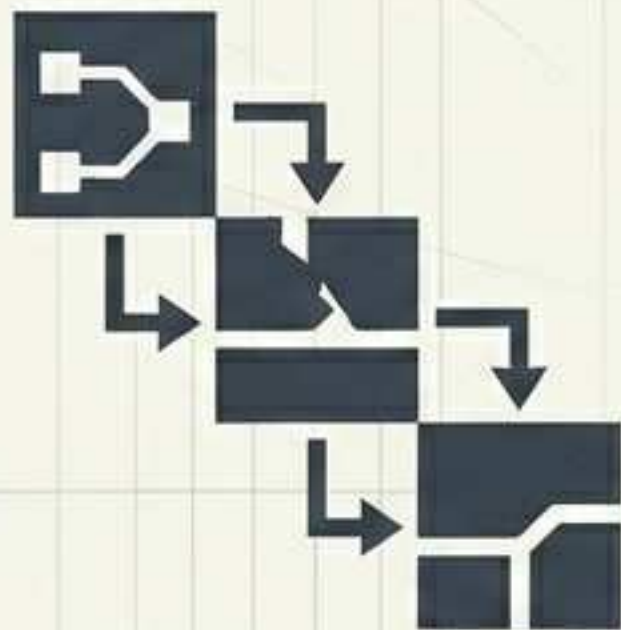
Execution is strictly irreversible. A world that could return to a prior state would undo history, which is not computation but its negation.

# Types are Historical Proofs, Not Value Sets



- In a history-primary setting, a value is not merely a member of a set. It is the result of a path.
- A type is a mathematical proof (certificate) that the history producing the value did not violate any constraints.
- Implementation requires the Calculus of Constructions (CoC) so types can depend directly on the historical terms that produced them.

# Rethinking the Computational Substrate



## Compilers (Event IR)

Correctness is no longer just output equivalence. It requires Replay Equivalence—the compiled VM must construct the exact same history as the interpreter.



## Garbage Collection

GC is not structural reference counting. It is admissibility-guided collapse, dropping dead history segments without breaking the type certificates of future continuations.



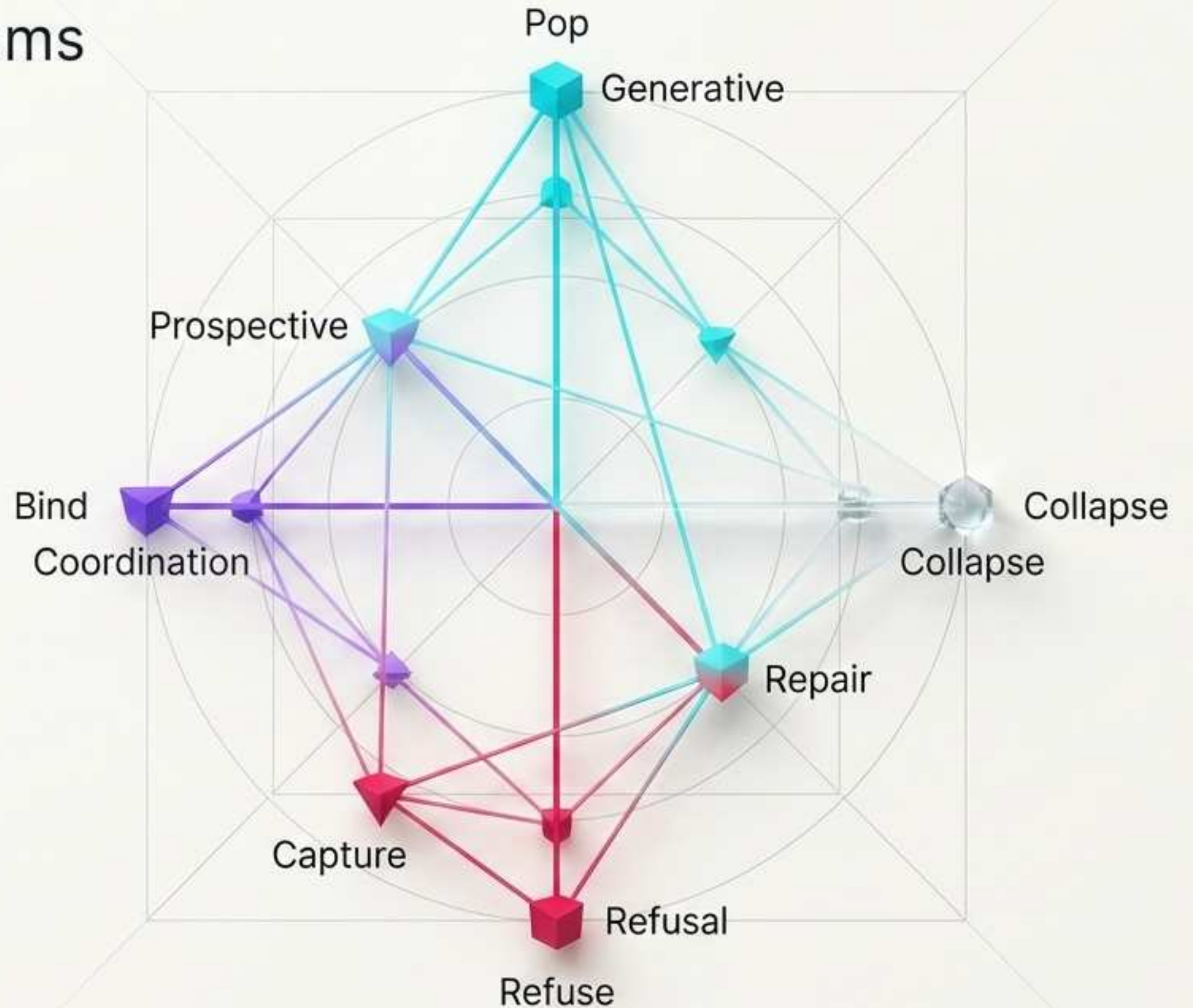
## Error Correction

Errors are not thrown exceptions. They are Certified Refusals built into the history with redundant paths, allowing recovery without erasing the failure record.

# Notes are Spherepop Programs

**Every note is a Spherepop program.  
Every Spherepop program is a note.**

Human note-taking maps perfectly  
onto the four mathematical operators  
of possibility space.



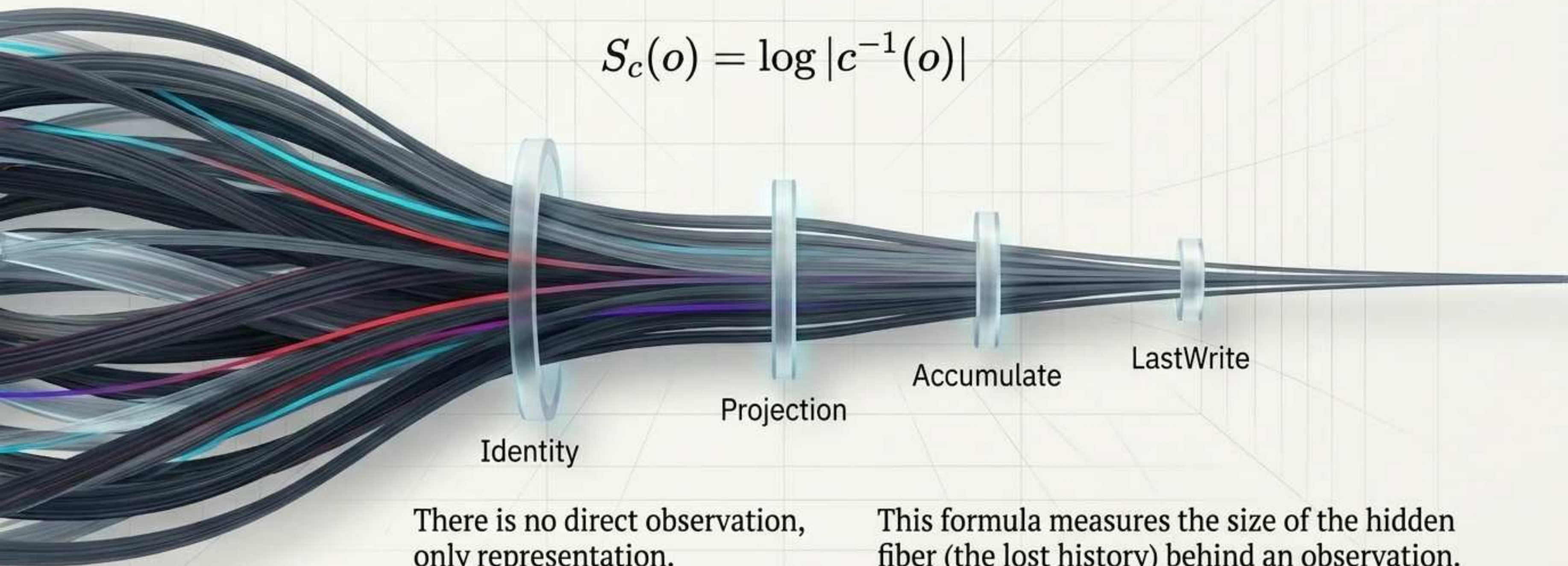
# The Ephemerality Inversion

| Y-Axis | Note Discarded | A discarded grocery list at the store exit.<br>100% Successful Note - Optimal |
|--------|----------------|-------------------------------------------------------------------------------|
|        | Note Preserved | A perfectly preserved, unread manuscript.<br>Unrealized Potential             |
|        |                | X-Axis                                                                        |
|        |                | Continuation Completed      Continuation Unfinished                           |

**The Inversion:** Ephemerality after completion is optimization, not failure. The celebrated surviving documents of antiquity survived precisely because their continuations were never completed.

# Measuring the Cost of Observation

$$S_c(o) = \log |c^{-1}(o)|$$



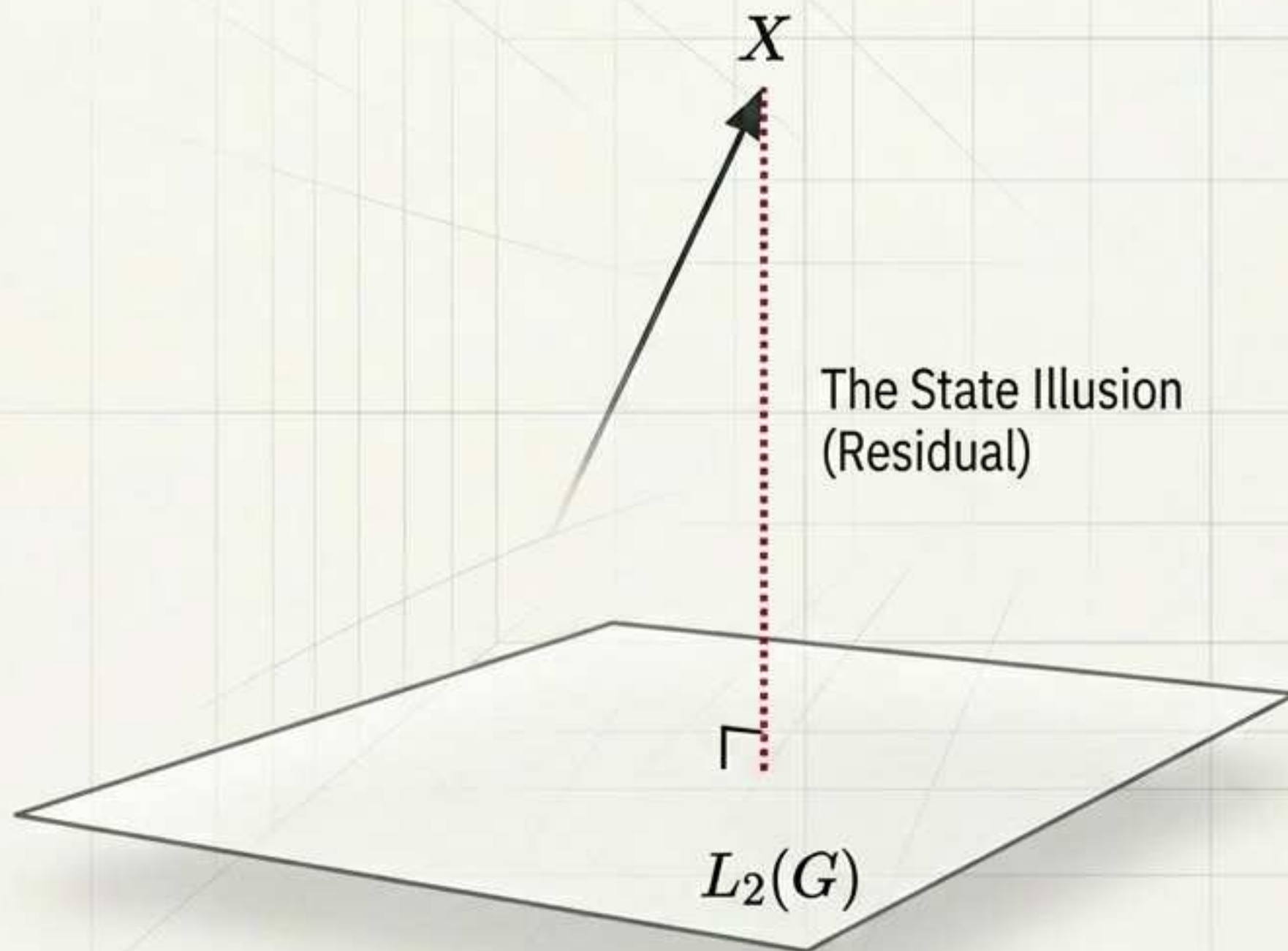
There is no direct observation,  
only representation.

Finer rules preserve more history  
but cost more to reconstruct.

This formula measures the size of the hidden  
fiber (the lost history) behind an observation.

Notes act as right-adjoints, lowering this  
reconstruction entropy.

# Continuous Math for Discrete Logic



## Discrete Logic

(Spherepop Collapse) is mathematically identical to

## Continuous Math

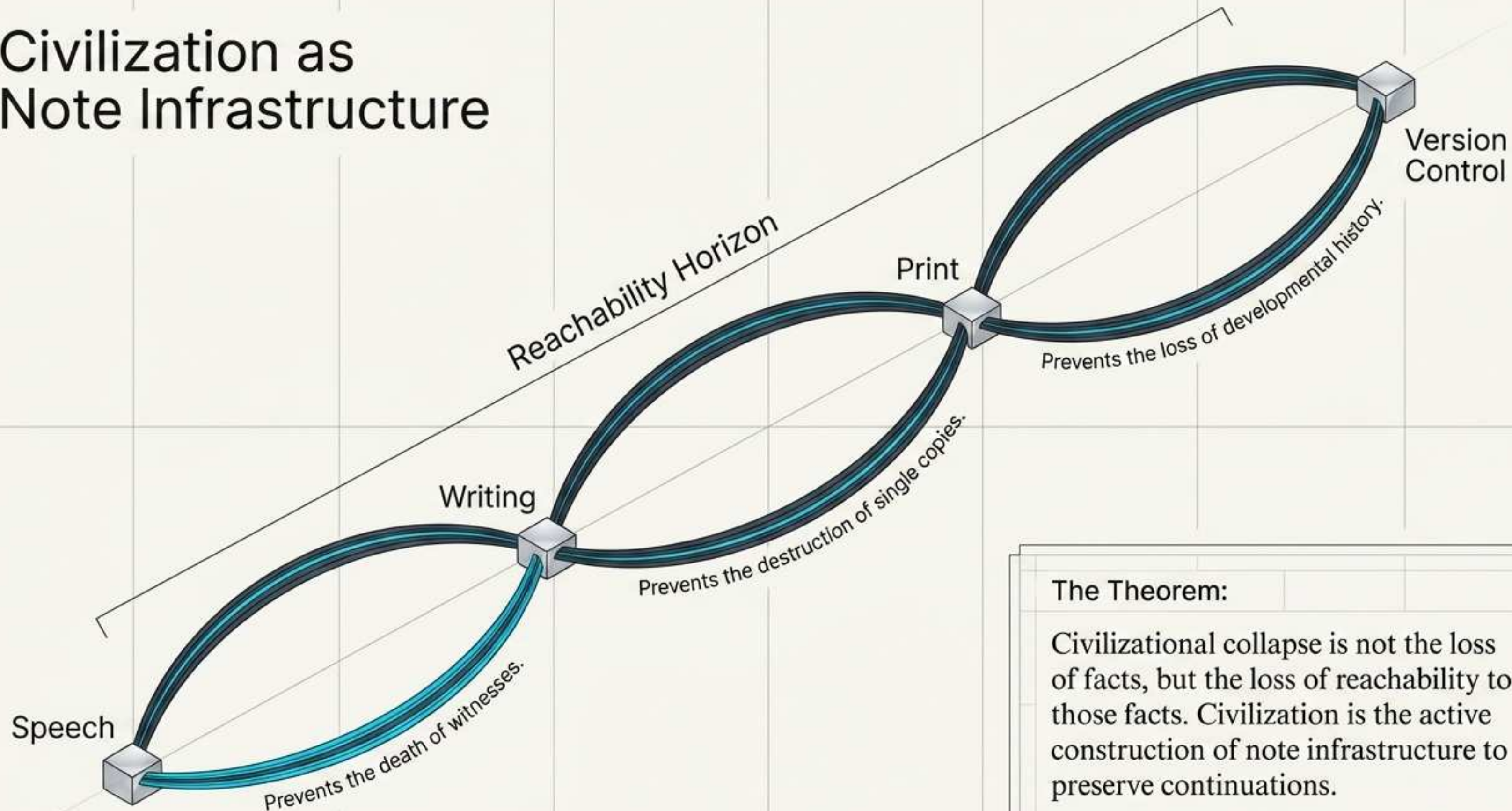
(Kolmogorov Conditional Expectation).

**Observational Entropy**  
(Discrete) maps perfectly to

**Variance**  
(Continuous).

The discarded history component is completely orthogonal to the observational framework, framework, making it mathematically invisible to the observer.

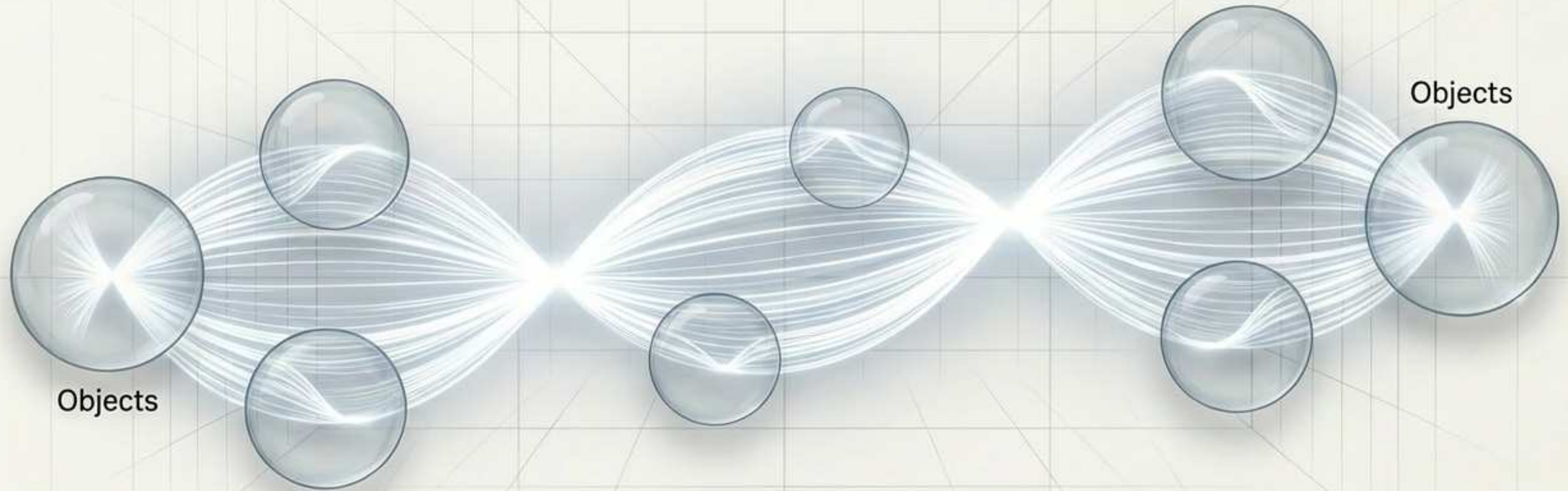
# Civilization as Note Infrastructure



## The Theorem:

Civilizational collapse is not the loss of facts, but the loss of reachability to those facts. Civilization is the active construction of note infrastructure to preserve continuations.

# Reality is Reachable History



The Noun Fallacy: Objects do not fundamentally exist; they are merely the stable quotients of underlying histories. We mistake the noun for the process.

Reality is not a collection of primitive objects. Reality is a structured space of reachable histories.

Conclusion: Objects are what histories look like after sufficient collapse. Spherepop gives us the algebra to reverse it.